ThaliBento

थाली + 弁当 · one-page summary

Thapar Japan Hackathon 2026 · Track JAL-PS-01 (Japan Airlines)
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Where we are: idea deck v5 submitted 18 Oct and shortlisted → **offline final 24–25 Oct at TIET**. We build a working prototype and pitch it to the judges and JAL mentors.

1 THE PROBLEM

JAL asks: **how can airlines understand and serve the food needs of Indian passengers, including last-minute changes?** Indian diets are very varied, but airlines squeeze them into 4-letter meal codes, and a special-meal request is passed along by hand 5 times. It can break at every step:

1 Booking	2 Changes	3 Kitchen	4 Loading	5 Serving
Wrong code: an agent books HNML (non-veg) for a vegetarian; nothing checks it	Request lost after a rebooking or codeshare	Counts mixed up: 28 non-veg instead of 28 veg	Meal missing on board	Veg passenger served non-veg

81% of Indians avoid some kind of meat · 315,000 Indians visited Japan in 2025 (+35%) · nobody publishes how often special meals fail.

2 OUR SOLUTION

ThaliBento is a safety net that plugs into JAL's existing systems. It **catches lost or wrong meals before they reach the seat**, and **finds a safe meal on board** when a passenger asks late. Example: Priya, vegetarian with no onion or garlic, flying IndiGo → JAL Delhi–Tokyo.

BEFORE THE JOURNEY	AT THE CATERER	AT THE AIRPORT	ON BOARD
Fills a short form once → diet rules → JAL code + kitchen note	Clear label; QR looks up her seat on the day	Lost request caught; staff fix it before the kitchen deadline	One galley check in the crew iPad app; backup if missing

3 FOUR THINGS NO AIRLINE DOES TODAY

Lost-meal alerts
Reads JAL's passenger lists and spots special meals dropped after a rebooking or codeshare. Staff approve every fix.

Record of what went wrong
5 checkpoints per meal. The log shows which step broke, so JAL finally gets real failure numbers.

Late requests matched on board
Asked at check-in or gate? Try a spare special meal → main-menu veg → sealed backup → lounge voucher.

Hidden-ingredient check
AI reads Japanese recipe sheets (dashi = fish stock). Our diet rules decide what's safe.

4 WHY JAL COULD USE IT

- **Read-only:** never edits JAL's booking system
- **No new crew app:** extra screen in i-PAXINFO, their iPad app
- **Same kitchens**, clearer labels, one backup veg meal
- **Rules decide, AI only reads** menus
- **Food rules, not religion;** opt-in, deleted after the trip

5 TECH WE BUILD WITH

Python rules engine (diets as JSON rules) · passenger-list parser to spot dropped meals · **Google OR-Tools** for late-request matching · FastAPI + Postgres · React screens + offline iPad screen · demo on a made-up JL740 Delhi–Tokyo flight.

6 THE 5-MINUTE DEMO

1. Priya fills the form → gets AVML + "no onion or garlic" note
2. Rebooked at 23:00 → alert fires → staff request her meal again
3. Caterer prints her label; recipe check flags dashi in miso soup
4. Galley check finds her meal; a Jain meal is missing → backup used
5. Jain request at the gate → matcher finds a safe spare on board

7 HOW WE'RE JUDGED

30% Technical	20% Problem	20% Innovation	20% Feasibility	10% Impact

Prize pool ₹1.1 lakh · 1st ₹40k · 2nd ₹30k · 3rd ₹20k

8 TO DO NOW

- [ ] Write PLAN.md in the code repo; everyone does laptop setup
- [ ] Build: shared schema → rules engine → list parser + alerts → **late-request matcher** (centrepiece) → galley screen
- [ ] Record a backup demo video on day 1
- [ ] Get one real number: quick flyer survey, chat with crew / caterer

9 THE PILOT WE PITCH

Delhi → Tokyo, 3 months, read-only.

- **Month 1 · Measure:** count lost meals, and where
- **Month 2 · Alert:** staff fix lost requests in time
- **Month 3 · Check on board:** labels, galley check, late matching

What JAL gets: fewer "wrong meal" complaints (courts award ₹40–65k), real failure data, works on any route.